

CY1308 Physics



LECTURE 6

DYNAMICS OF ROTATIONAL MOTION (Y&F CH 9,10)

Angular momentum of a particle

Suppose a particle of mass m has momentum $\mathbf{p}$ and position vector $\mathbf{r}$ with respect to the origin O in some chosen inertial reference frame.

Then the general definition of **angular momentum $\mathbf{L}$** , of the particle about point O is the cross product,

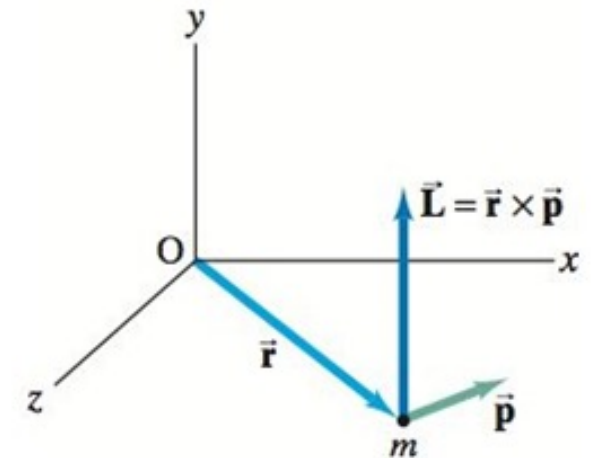
$$\vec{L} = \vec{r} \times \vec{p}$$

It is a vector, given by the direction of the cross product, and its direction is perpendicular to both $\mathbf{r}$ and $\mathbf{p}$.

Its magnitude is given by

$$L = rp \sin \theta$$

where θ is the angle between the position vector $\mathbf{r}$ and momentum $\mathbf{p}$.



The angular momentum of a particle of mass m is given by $\vec{L} = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$.

Angular momentum of a rigid body

For a rigid body rotating around an axis of symmetry, the angular momentum is the sum of angular momenta of each i-th mass element.

For simplicity, consider that the body is rotating about the z-axis with angular velocity ω . The angular momentum of the body is

$$\begin{aligned}\vec{L} &= \sum_i \vec{r}_i \times \vec{p}_i = \sum_i \vec{r}_i \times (\Delta m_i \vec{v}_i) \\ &= \sum_i (\Delta m_i r_i v_i \sin \theta_i) \hat{u}_z = \sum_i (\Delta m_i r_i^2 \sin^2 \theta_i) (\omega \hat{u}_z)\end{aligned}$$

$$\therefore \vec{L} = I \vec{\omega}$$

This can be generalized to the formula for rotation about a symmetry (principal)* axis of a rigid body:

Angular momentum of a rigid body rotating around a symmetry axis	$\vec{L} = I \vec{\omega}$	Moment of inertia of rigid body about symmetry axis Angular velocity vector of rigid body
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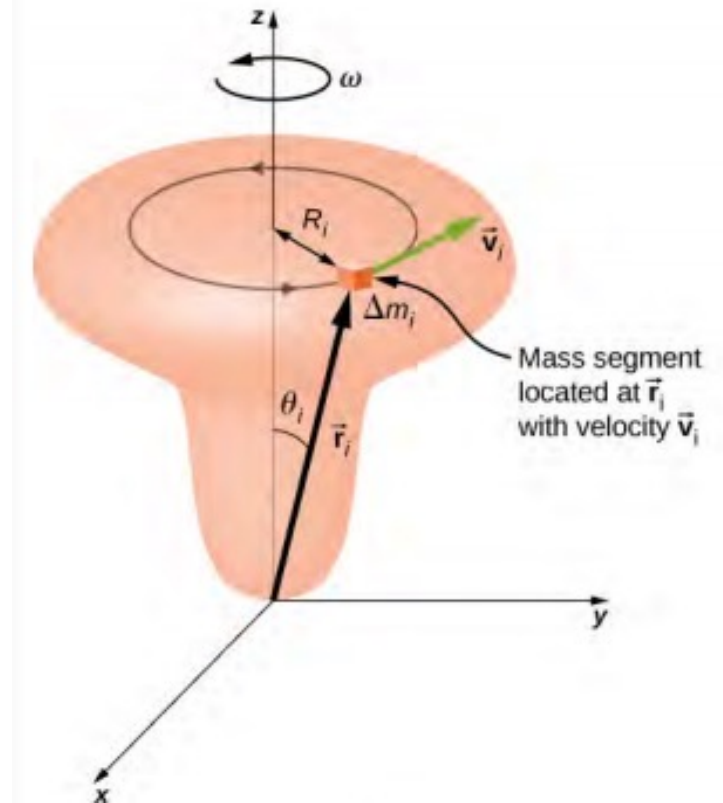


Image source: Libretexts

Angular momentum and torque

Angular momentum of a rigid body rotating around a symmetry axis

$$\vec{L} = I\vec{\omega}$$

Moment of inertia of rigid body about symmetry axis

Angular velocity vector of rigid body

If the axis is NOT an axis of symmetry, or NOT parallel to angular velocity, the above relation holds but the moment of inertia is written as a 3x3 matrix.

For any system of particles, the rate of change of the total angular momentum equals the sum of the torques of all forces acting on all the particles:

For a system of particles:

Sum of external torques on the system

$$\sum \vec{\tau} = \frac{d\vec{L}}{dt}$$

Rate of change of total angular momentum $\vec{L}$ of system

Note that this is only valid when the net external torque (LHS) and angular momentum $\vec{L}$, are calculated with respect to either

1. The origin of an inertial reference frame, or
2. The centre of mass of a system of particles or rigid body. **(Important!)**

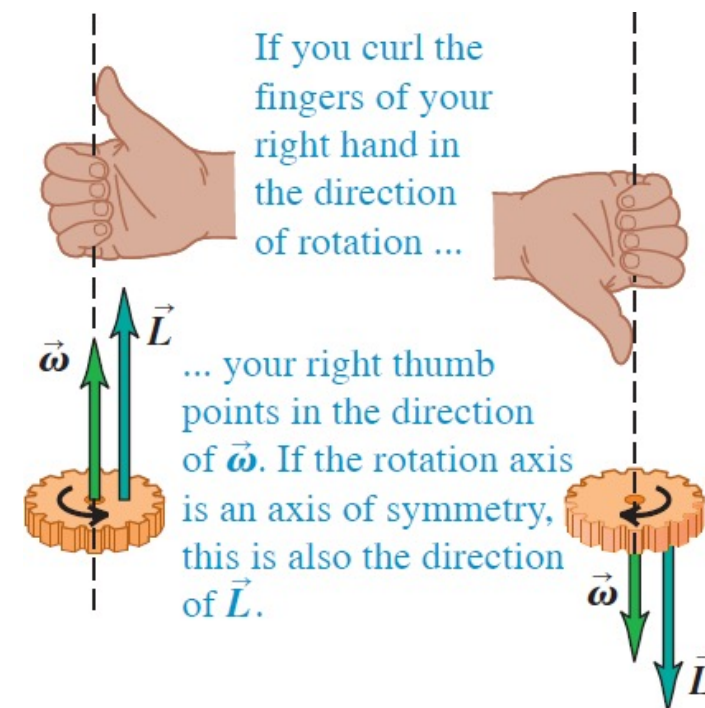


Image source: Young and Freedman

Torque and angular momentum about CM – derivation (1)

Suppose the rigid body is accelerating. Then the position and momentum of the i -th mass element is

$$\vec{\mathbf{r}}_i = \vec{\mathbf{r}}_{\text{CM}} + \vec{\mathbf{r}}_i^*.$$

If we multiply each term by m_i and take the derivative of this equation, we can write

$$\vec{\mathbf{p}}_i = m_i \frac{d\vec{\mathbf{r}}_i}{dt} = m_i \frac{d}{dt}(\vec{\mathbf{r}}_i^* + \vec{\mathbf{r}}_{\text{CM}}) = m_i \vec{\mathbf{v}}_i^* + m_i \vec{\mathbf{v}}_{\text{CM}} = \vec{\mathbf{p}}_i^* + m_i \vec{\mathbf{v}}_{\text{CM}}.$$

The angular momentum with respect to the CM is

$$\vec{\mathbf{L}}_{\text{CM}} = \sum_i (\vec{\mathbf{r}}_i^* \times \vec{\mathbf{p}}_i^*) = \sum_i \vec{\mathbf{r}}_i^* \times (\vec{\mathbf{p}}_i - m_i \vec{\mathbf{v}}_{\text{CM}}).$$

Then, taking the time derivative, we have

$$\frac{d\vec{\mathbf{L}}_{\text{CM}}}{dt} = \sum_i \left(\frac{d\vec{\mathbf{r}}_i^*}{dt} \times \vec{\mathbf{p}}_i^* \right) + \sum_i \left(\vec{\mathbf{r}}_i^* \times \frac{d\vec{\mathbf{p}}_i^*}{dt} \right).$$

The first term on the right is $\vec{\mathbf{v}}_i^* \times m_i \vec{\mathbf{v}}_i^*$ and equals zero because $\vec{\mathbf{v}}_i^*$ is parallel to itself ($\sin \theta = 0$). Thus

$$\begin{aligned} \frac{d\vec{\mathbf{L}}_{\text{CM}}}{dt} &= \sum_i \vec{\mathbf{r}}_i^* \times \frac{d}{dt}(\vec{\mathbf{p}}_i - m_i \vec{\mathbf{v}}_{\text{CM}}) \\ &= \sum_i \vec{\mathbf{r}}_i^* \times \frac{d\vec{\mathbf{p}}_i}{dt} - \left(\sum_i m_i \vec{\mathbf{r}}_i^* \right) \times \frac{d\vec{\mathbf{v}}_{\text{CM}}}{dt}. \end{aligned}$$

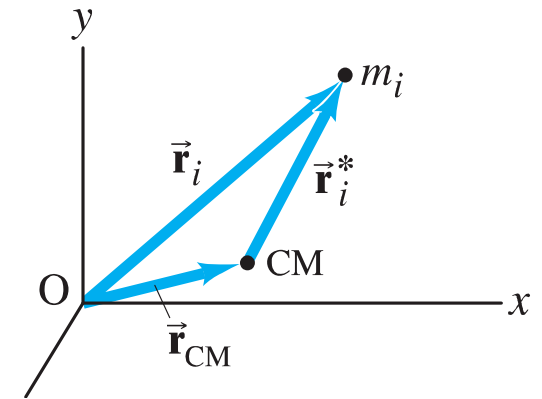


FIGURE 11-14 The position of m_i in the inertial frame is $\vec{\mathbf{r}}_i$; with regard to the CM (which could be accelerating) it is $\vec{\mathbf{r}}_i^*$, where $\vec{\mathbf{r}}_i = \vec{\mathbf{r}}_i^* + \vec{\mathbf{r}}_{\text{CM}}$ and $\vec{\mathbf{r}}_{\text{CM}}$ is the position of the CM in the inertial frame.

Image source: Young and Freedman

Torque and angular momentum about CM – derivation (2)

$$\begin{aligned}\frac{d\vec{L}_{\text{CM}}}{dt} &= \sum_i \vec{r}_i^* \times \frac{d}{dt} (\vec{p}_i - m_i \vec{v}_{\text{CM}}) \\ &= \sum_i \vec{r}_i^* \times \frac{d\vec{p}_i}{dt} - \left(\sum_i m_i \vec{r}_i^* \right) \times \frac{d\vec{v}_{\text{CM}}}{dt}.\end{aligned}$$

The second term on the right is zero since, $\sum m_i \vec{r}_i^* = M \vec{r}_{\text{CM}}^*$, and $\vec{r}_{\text{CM}}^* = 0$ by definition (the position of the CM is at the origin of the CM reference frame). Furthermore, by Newton's second law we have

$$\frac{d\vec{p}_i}{dt} = \vec{F}_i,$$

where $\vec{F}_i$ is the net force on m_i . (Note that $d\vec{p}_i^*/dt \neq \vec{F}_i$ because the CM may be accelerating and Newton's second law does not hold in a noninertial reference frame.) Consequently

$$\frac{d\vec{L}_{\text{CM}}}{dt} = \sum_i \vec{r}_i^* \times \vec{F}_i = \sum_i (\vec{\tau}_i)_{\text{CM}} = \sum \vec{\tau}_{\text{CM}},$$

Therefore:
$$\frac{d\vec{L}_{\text{CM}}}{dt} = \sum \vec{\tau}_{\text{CM}}. \quad [\text{even if accelerating}]$$

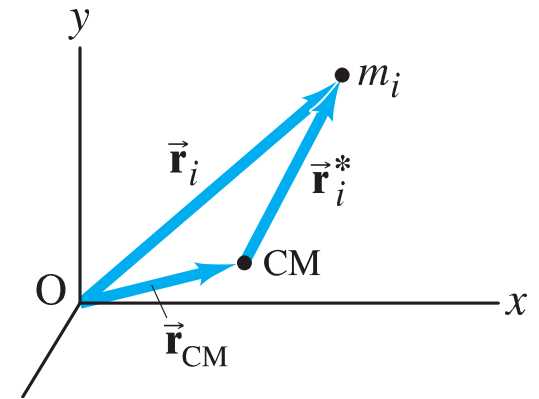


FIGURE 11-14 The position of m_i in the inertial frame is $\vec{r}_i$; with regard to the CM (which could be accelerating) it is $\vec{r}_i^*$, where $\vec{r}_i = \vec{r}_i^* + \vec{r}_{\text{CM}}$ and $\vec{r}_{\text{CM}}$ is the position of the CM in the inertial frame.

Image source: Young and Freedman

Conservation of angular momentum

When the net external torque acting on a system is zero, the total angular momentum of the system is constant (conserved).

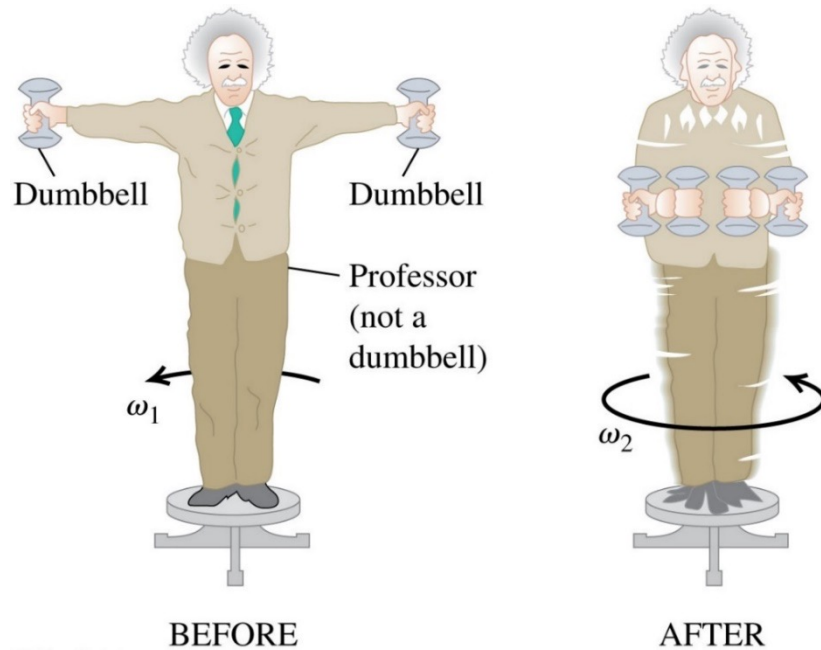


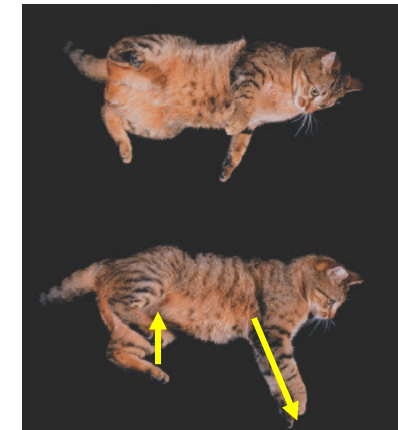
Image source: Young and Freedman



A falling cat twists different parts of its body in different directions so that it lands feet first. At all times during this process the angular momentum of the cat as a whole remains zero.

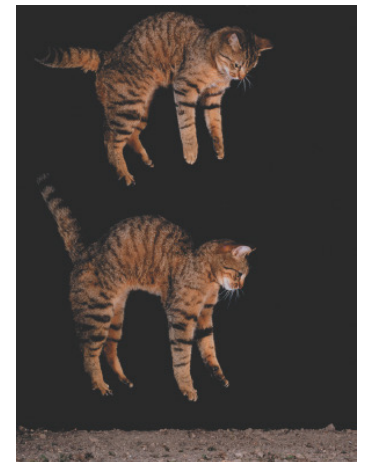
<https://youtu.be/RtWbpyjJqrU>

Slow Motion Flipping Cat Physics |
Smarter Every Day 58



By extending its front paws, the cat increases the MI of the front torso, and slows down the angular velocity of its front. Tucking in the rear paws has the reverse effect.

At the upright orientation, all paws extend to slow down any further rotation. Landing takes place over a long time to reduce the force of impact.



Video: conservation of angular momentum (1)

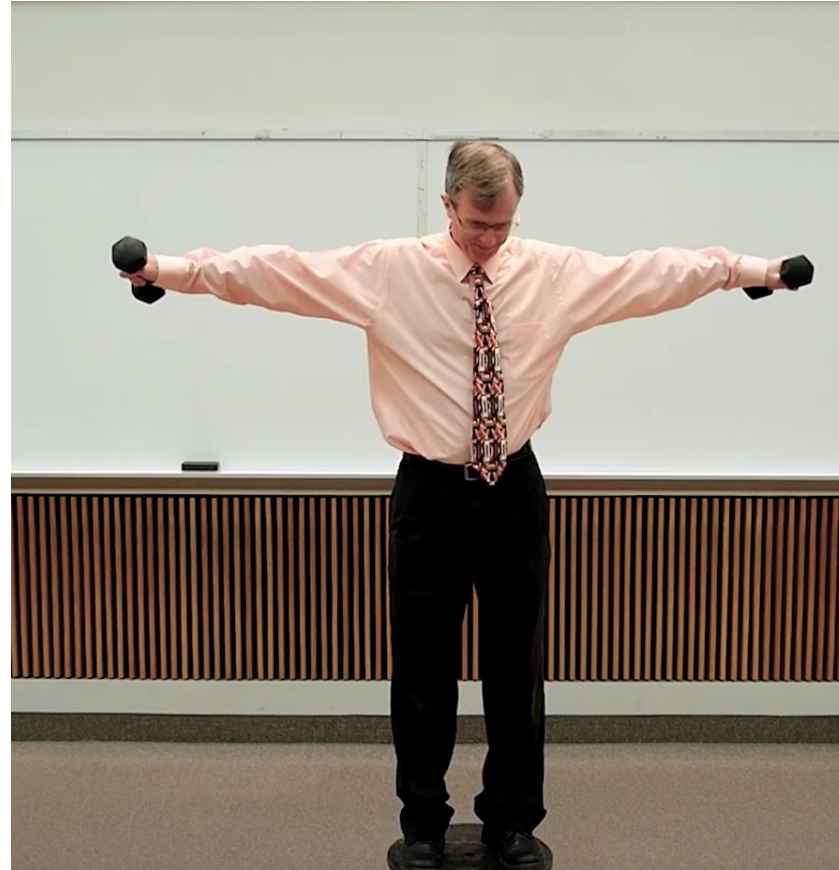
angular momentum

$$L = I\omega,$$

in the absence of a net
external *torque*

$$\frac{dL}{dt} = 0 \quad \text{and}$$

$$L = I\omega = \text{constant.}$$



<https://youtu.be/fwAjPOR28J0>

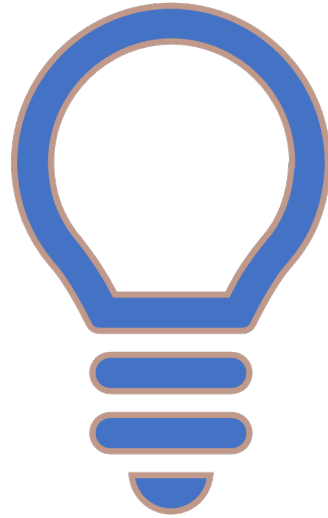
Video: conservation of angular momentum (2)

A man holding a spinning wheel stands on a stationary frictionless turntable. What will happen if he suddenly flips the bicycle wheel over so that it is spinning in the opposite direction?



Video:

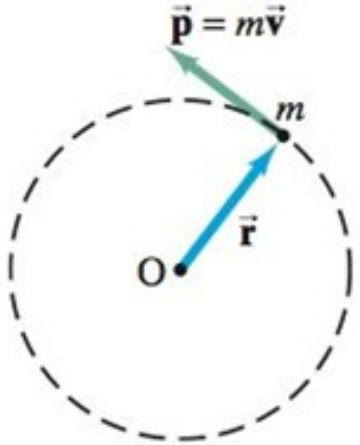




Worked Examples

Example 1: angular momentum of a particle

What is the angular momentum of the particle moving in a circle of radius r with speed v as shown in the figure?



Answer:

The value of the angular momentum $\mathbf{L}$ will depend on the choice of the origin O .

Calculation of $\mathbf{L}$ about the centre of the circle (the choice of origin O) gives

$$L = r p \sin 90^\circ = mvr = mr^2\omega \quad (\text{magnitude})$$

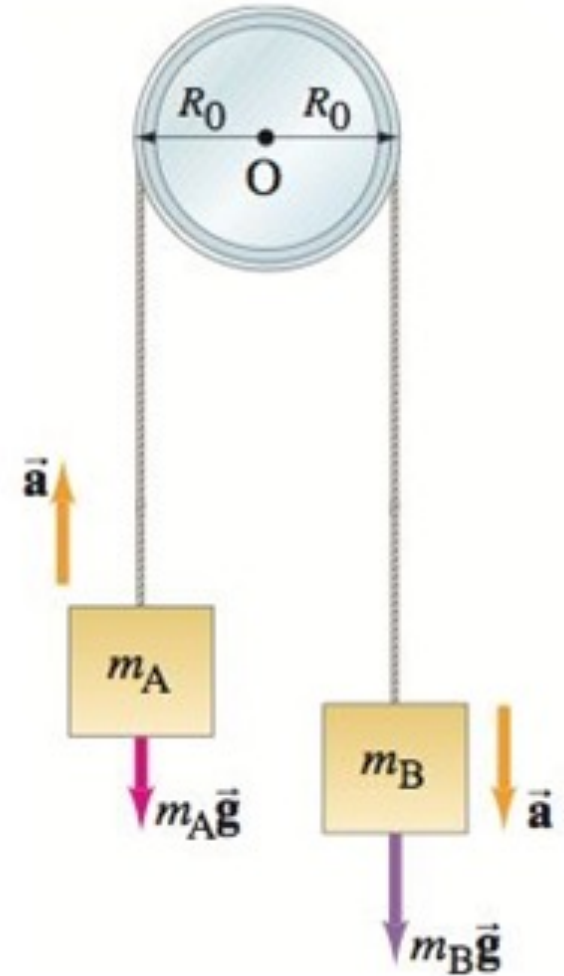
Pointing in the direction perpendicular to the plane of the circle, outwards towards the viewer.

For a single particle, $I = mr^2$. So $L = I \omega$.

Note that one could use in the derivation: $\vec{\omega} = (\vec{r} \times \vec{v}) / r^2$

Example 2: Pulley with mass

Atwood's machine. An *Atwood machine* consists of two masses, m_A and m_B , which are connected by an inelastic cord of negligible mass that passes over a pulley, Fig. 11–16. If the pulley has radius R_0 and moment of inertia I about its axle, determine the acceleration of the masses m_A and m_B , and compare to the situation where the moment of inertia of the pulley is ignored.



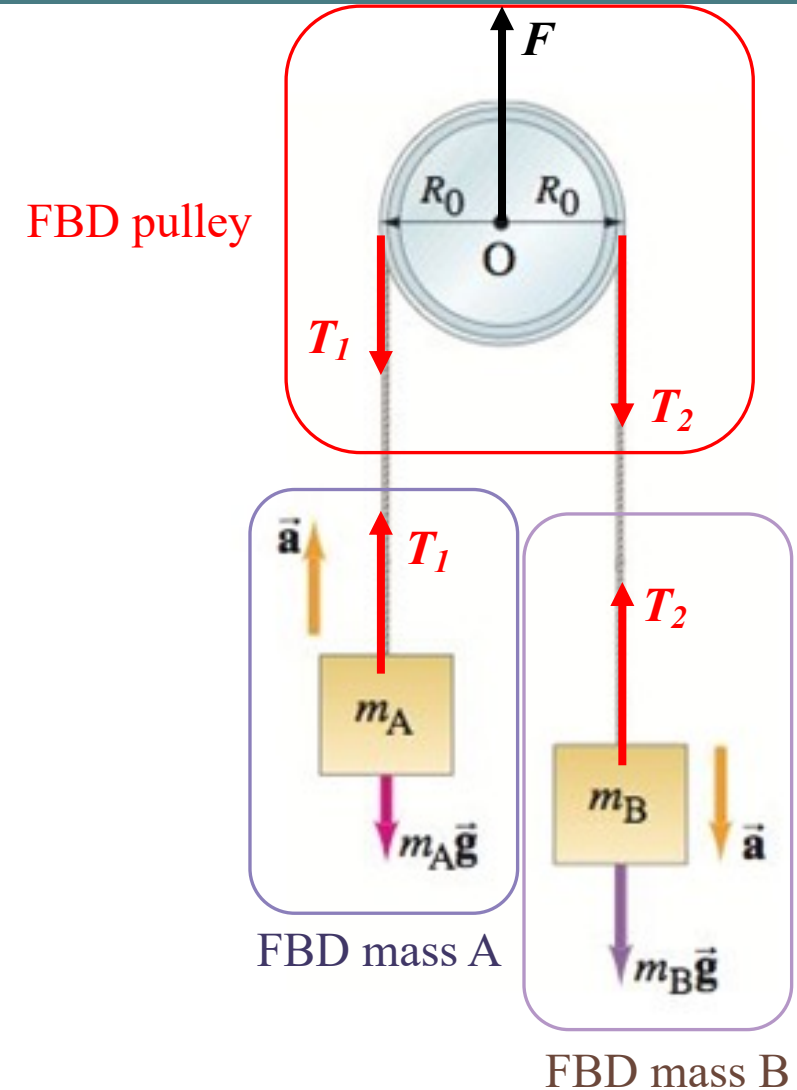
Source: Giancoli

Example 2: Pulley with mass (explanation)

The difference now (from previous lectures) is that pulley has mass and moment of inertia. It takes a torque to change its angular momentum, i.e. need a torque to start it rotating from rest.

We assume the following:

- (1) The rope does not slip on the pulley surface, so all the torque from the tensions goes to changing the angular momentum of the pulley instead of overcoming kinetic friction from the pulley surface. (Here, the implicit assumption is that there is static friction between the points of contact of the rope and the pulley surface. It is this static friction that gives rise to the different tensions on either sides. Read further – Capstan equation.)
- (2) There is no frictional torque from the axle of the pulley, so there are no other torques on the pulley apart from that arising from tensions.
- (3) The rope has no mass, i.e. no force is needed to accelerate it.



Example 2: Pulley with mass (solution)

Solution:

Ex 2.

$a = g \left(\frac{m_B - m_A}{m_B + m_A + \frac{I_P}{R_0^2}} \right)$; $T_1 = m_A g \left(\frac{2m_B R_0^2 + I_P}{m_A R_0^2 + m_B R_0^2 + I_P} \right)$; $T_2 = m_B g \left(\frac{2m_A R_0^2 + I_P}{m_A R_0^2 + m_B R_0^2 + I_P} \right)$

METHOD 1

$T_1 - m_A g = m_A a$
 $m_B g - T_2 = m_B a$
 $T_2 R_0 - T_1 R_0 = I_P \alpha$
 No slipping of rope on pulley: $a = R_0 \alpha$

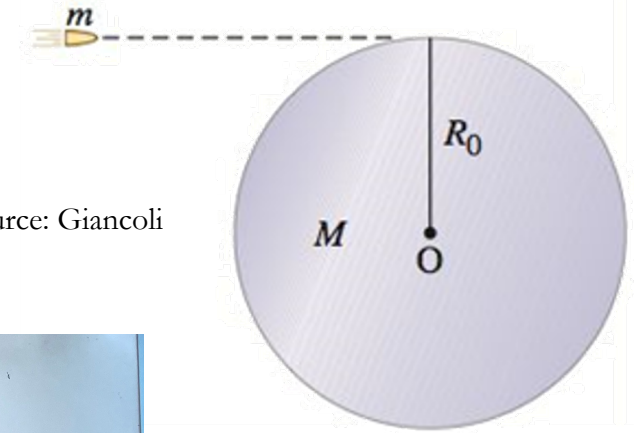
METHOD 2

Total angular momentum of system about P:
 $L_P = I_P \omega + m_B v R_0 + m_A v R_0$
 Net torque on system about horiz. axis thru P:
 $\tau_P = m_B g R_0 - m_A g R_0$
 Using $\tau_P = \frac{dL_P}{dt}$
 $(m_B - m_A) g R_0 = \left(m_A + m_B + \frac{I_P}{R_0^2} \right) R_0 \frac{dv}{dt}$
 $\therefore a = g \left(\frac{m_B - m_A}{m_A + m_B + \frac{I_P}{R_0^2}} \right)$

$\vec{L} = \vec{r} \times \vec{p}$

Example 3: Collision and angular momentum

Bullet strikes cylinder edge. A bullet of mass m moving with velocity v strikes and becomes embedded at the edge of a cylinder of mass M and radius R_0 , as shown in Fig. 11-22. The cylinder, initially at rest, begins to rotate about its symmetry axis, which remains fixed in position. Assuming no frictional torque, what is the angular velocity of the cylinder after this collision? Is kinetic energy conserved?



Source: Giancoli

Solution:

Ex 3 Top view

Before

After

Cons. of angular momentum

$$L_i = \vec{r} \times \vec{p} = \vec{r} \times m\vec{v}$$

About O:

$$L_i = mvR_0 \text{ (bullet)}$$

①

$$v_f = R_0 \omega_f$$

$$L_f = m v_f R_0 + I_{\text{disk}} \omega_f$$

bullet

$$L_f = (I_{\text{disk}} + m R_0^2) \omega_f$$

②

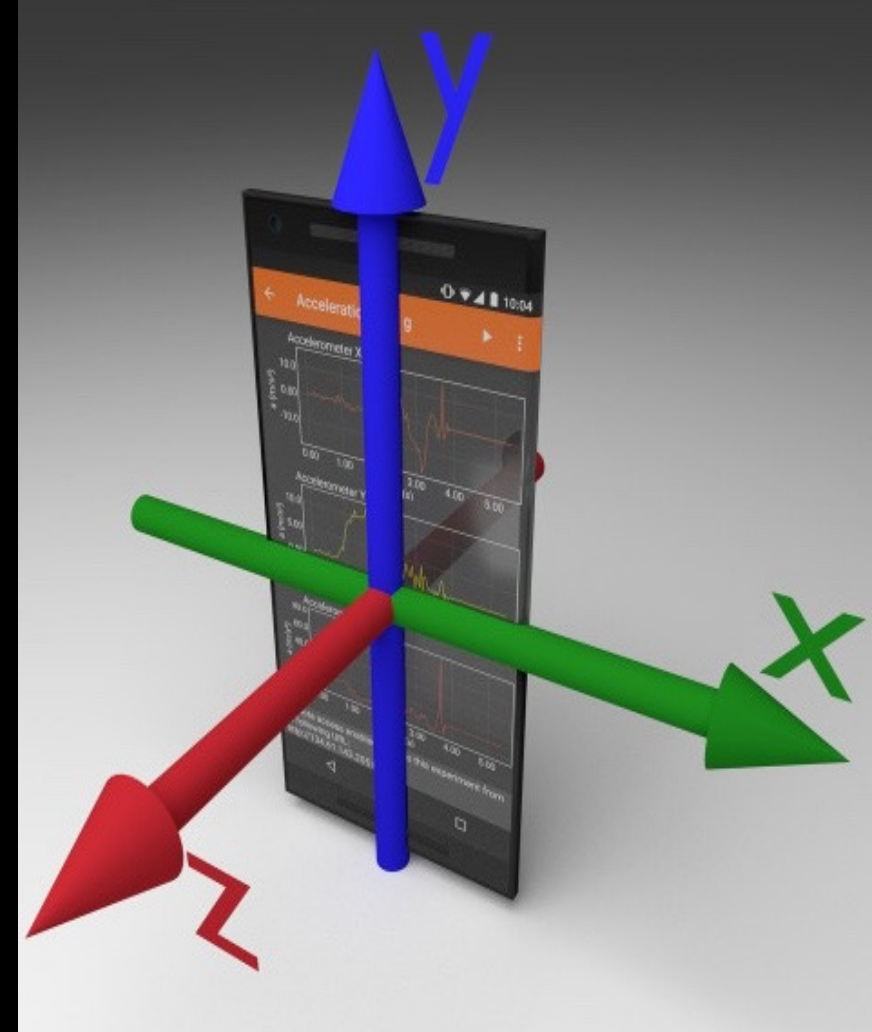
Initial KE, $K_i = \frac{1}{2} m v^2$

Final KE, $K_f = \frac{1}{2} m v_f^2 + \frac{1}{2} I_{\text{disk}} \omega_f^2$

$$= \frac{1}{2} m R_0^2 \omega_f^2 + \frac{1}{2} \left(\frac{1}{2} M R_0^2 \right) \omega_f^2$$

$$= \frac{1}{2} \left[\frac{m}{1 + \left(\frac{m}{2M} \right)} \right] v^2$$

$\therefore K_f < K_i$
KE not conserved.



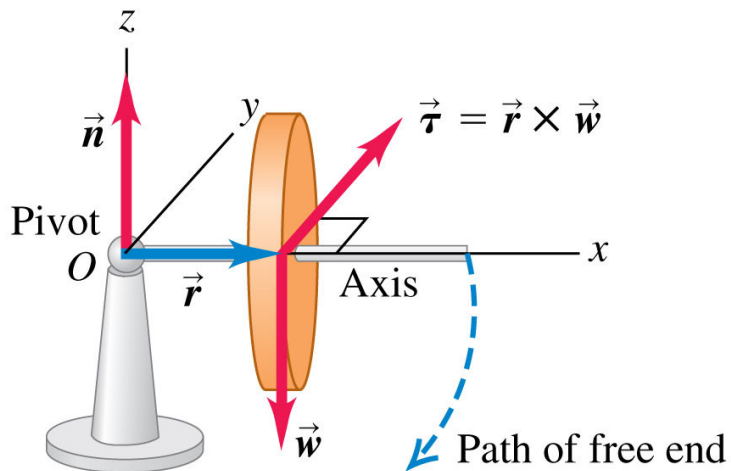
Gyroscopic motion

Gyroscopes, torque and precession (1)

If a flywheel is initially not spinning, its initial angular momentum is zero.

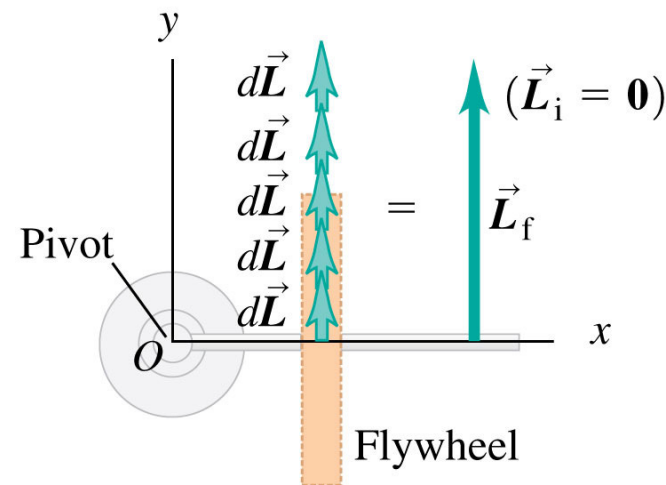
In each successive time interval dt , the torque produces a change in the angular momentum in the same direction as the torque, and the flywheel axis falls.

(a) Nonrotating flywheel falls



When the flywheel is not rotating, its weight creates a torque around the pivot, causing it to fall along a circular path until its axis rests on the table surface.

(b) View from above as flywheel falls



In falling, the flywheel rotates about the pivot and thus acquires an angular momentum $\vec{L}$. The *direction* of $\vec{L}$ stays constant.

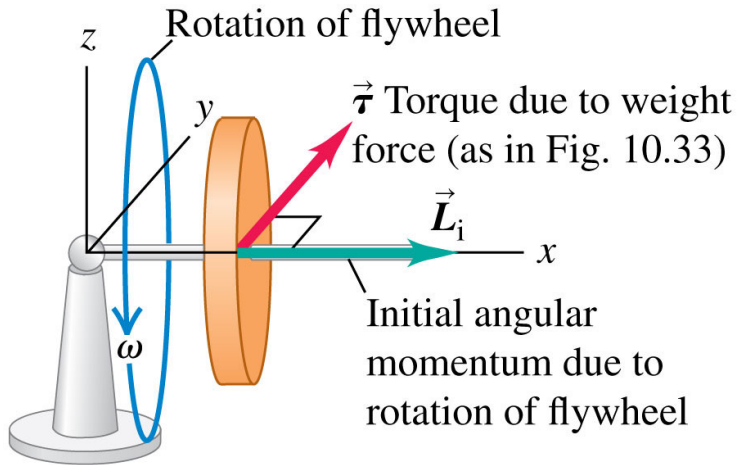
Image source: Young and Freedman

Gyroscopes, torque and precession (2)

For a gyroscope, the axis of rotation changes direction. The motion of this axis is called **precession**.

(a) Rotating flywheel

When the flywheel is rotating, the system starts with an angular momentum $\vec{L}_i$ parallel to the flywheel's axis of rotation.



This flywheel is initially spinning, with a large angular momentum.

(b) View from above

Now the effect of the torque is to cause the angular momentum to precess around the pivot. The gyroscope circles around its pivot without falling.

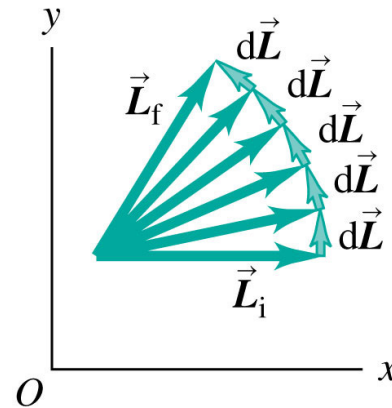


Image source: Young and Freedman

Because the initial angular momentum is not zero, each change in angular momentum is perpendicular to the angular momentum. As a result, the magnitude L remains the same but the angular momentum changes its direction continuously.

Derivation of precession rate of gyroscope

Change in angular momentum as a result of external torque due to weight is

$$d\vec{L} = \vec{\tau}_{\text{ext}} dt$$

We see that

$$dL = L \sin \theta d\phi$$

The angular velocity of precession is

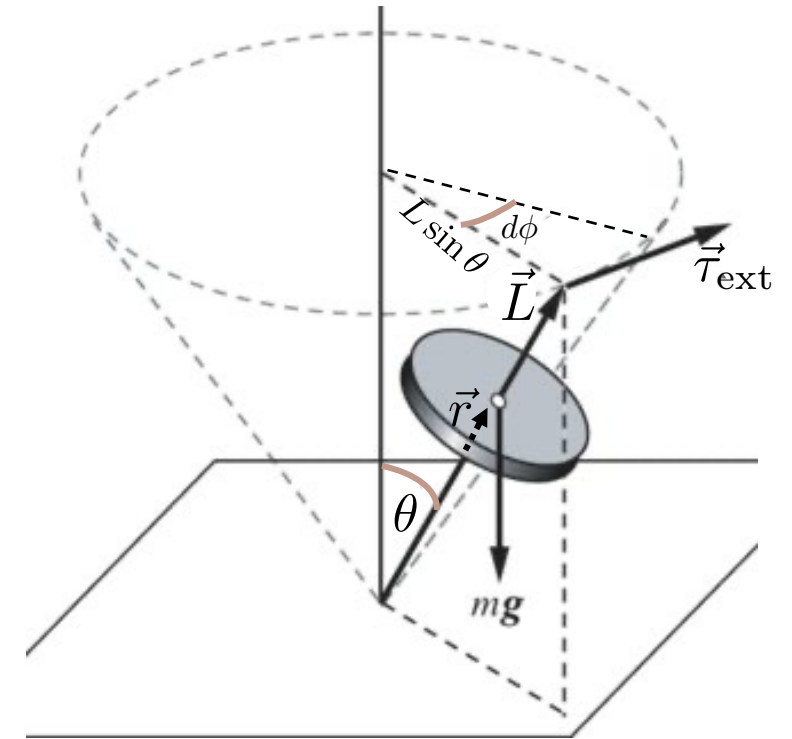
$$\Omega = d\phi/dt$$

$$\Rightarrow \Omega = \frac{1}{L \sin \theta} \frac{dL}{dt} = \frac{\tau_{\text{ext}}}{L \sin \theta} = \frac{Mgr \sin \theta}{L \sin \theta}$$

$$\therefore \Omega = \frac{Mgr}{L} = \frac{Mgr}{\underbrace{(kMR_0^2)}_{\text{Moment of Inertia } I} \omega} = \frac{gr}{kR_0^2 \omega}$$

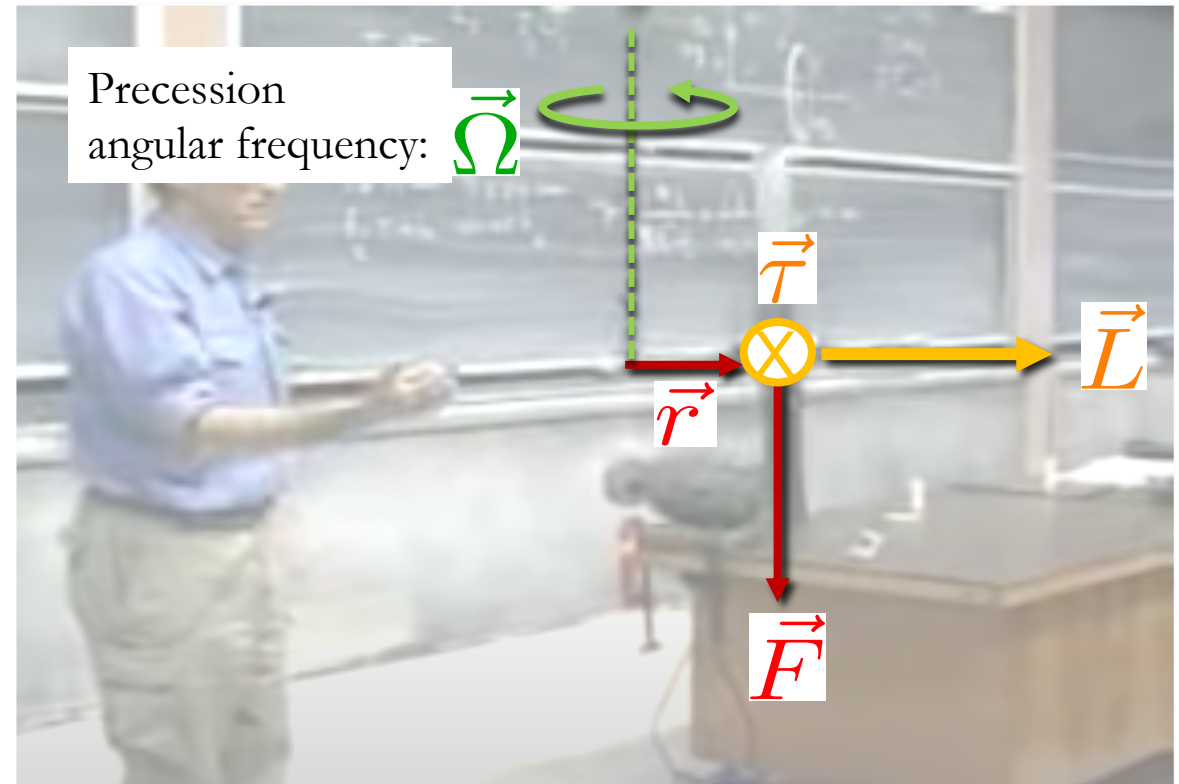
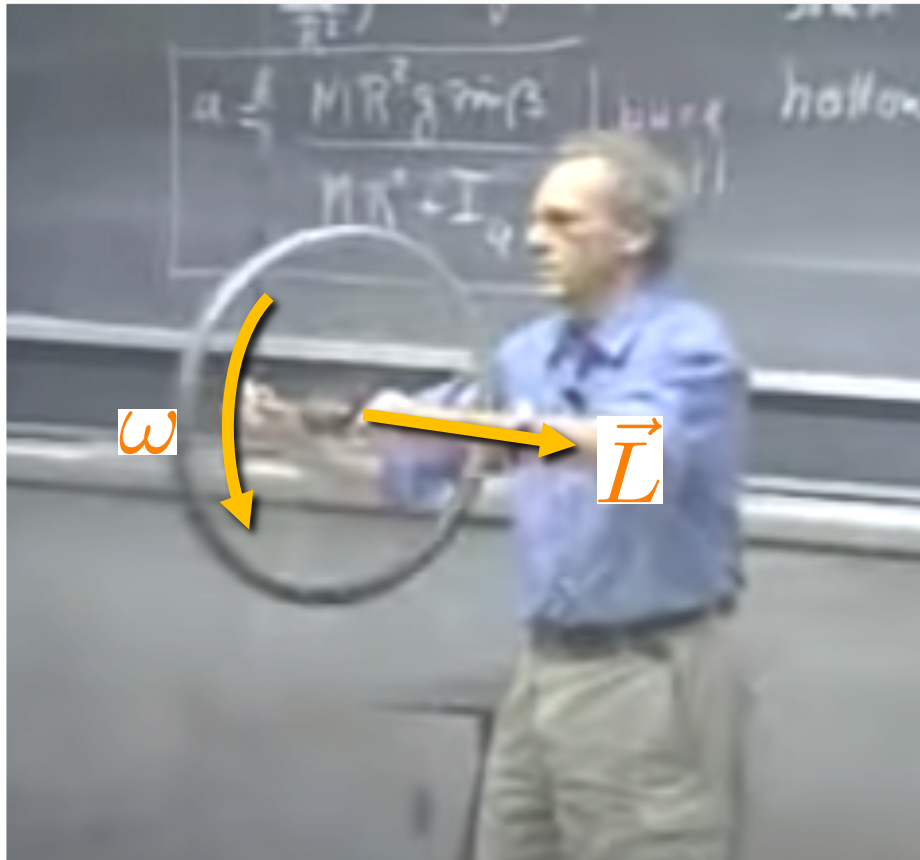
(where k is the dimensionless pre-factor of the MI, and R_0 is the radius of the flywheel)

which does not depend on the angle θ but is inversely proportional to the gyroscope's angular velocity ω , and proportional to the position of the CM from the pivot r .



Video: Gyroscopic precession

<https://youtu.be/NeXIV-wMVUk>



Precession caused by torque produced by the weight:

$$\vec{\tau} = \vec{r} \times \vec{F} = d\vec{L}/dt$$

Further watching and reading

Further reading:

- Young and Freedman Chapters 9, 10
- Dynamic Study Module in MasteringPhysics

Further watching:

- MEMS sensors in your mobile phone: <https://youtu.be/eqZgxR6eRjo>
- Spinning toys (Physics Girl):
<https://youtu.be/1Tx7FgZuV3U>

- **(Extra) Imbalanced rotation** - figure on right

In slide 4, we mentioned that $\vec{L} = I\vec{\omega}$ is not valid in general. An example is when the vectors $\vec{L}$, $\vec{\omega}$ are not parallel, as shown in this figure. This results in an imbalanced rotation, with forces on the bearings as a result of the imbalance.

- **(Extra) Capstan equation (Wikipedia):**
https://en.wikipedia.org/wiki/Capstan_equation
- (Extra) Gyroscopic nutation (Wikipedia):
<https://en.wikipedia.org/wiki/Nutation>

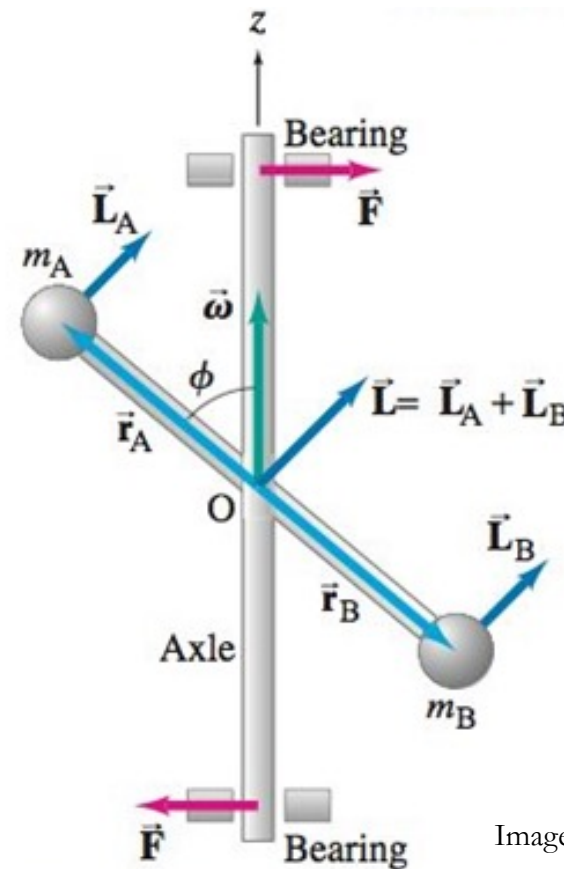


Image source: Giancoli

In this system $\vec{L}$ and $\vec{\omega}$ are not parallel. This is an example of rotational imbalance.